AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 15 · STUDENT LESSON

Confusion Matrix

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Read the reference. Read the prediction. Count the cell.

Build and interpret confusion counts without hiding error identities or axis conventions.

Keep one habit: Eight complete records → counting program → labelled twenty-case picture → complete Matrix Build → independent new counts.

MAIN LESSON · SECTION 1

What kind of mistake did the classifier make?

> Preserve the reference label, prediction and example ID before counting.

Learning goals

I can build a labelled **confusion matrix** from matched reference and **prediction** pairs.

I can distinguish **false positive**s from **false negative**s using a declared **positive class**.

I can check matrix totals and explain why equal accuracy can hide different errors.

Recall the saved classifier pairs

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

- Why should a reproducible run save **prediction**s linked to example IDs rather than only an accuracy score?

End goal

By the end, I can explain **confusion matrix** using a traced example and a stated limitation.

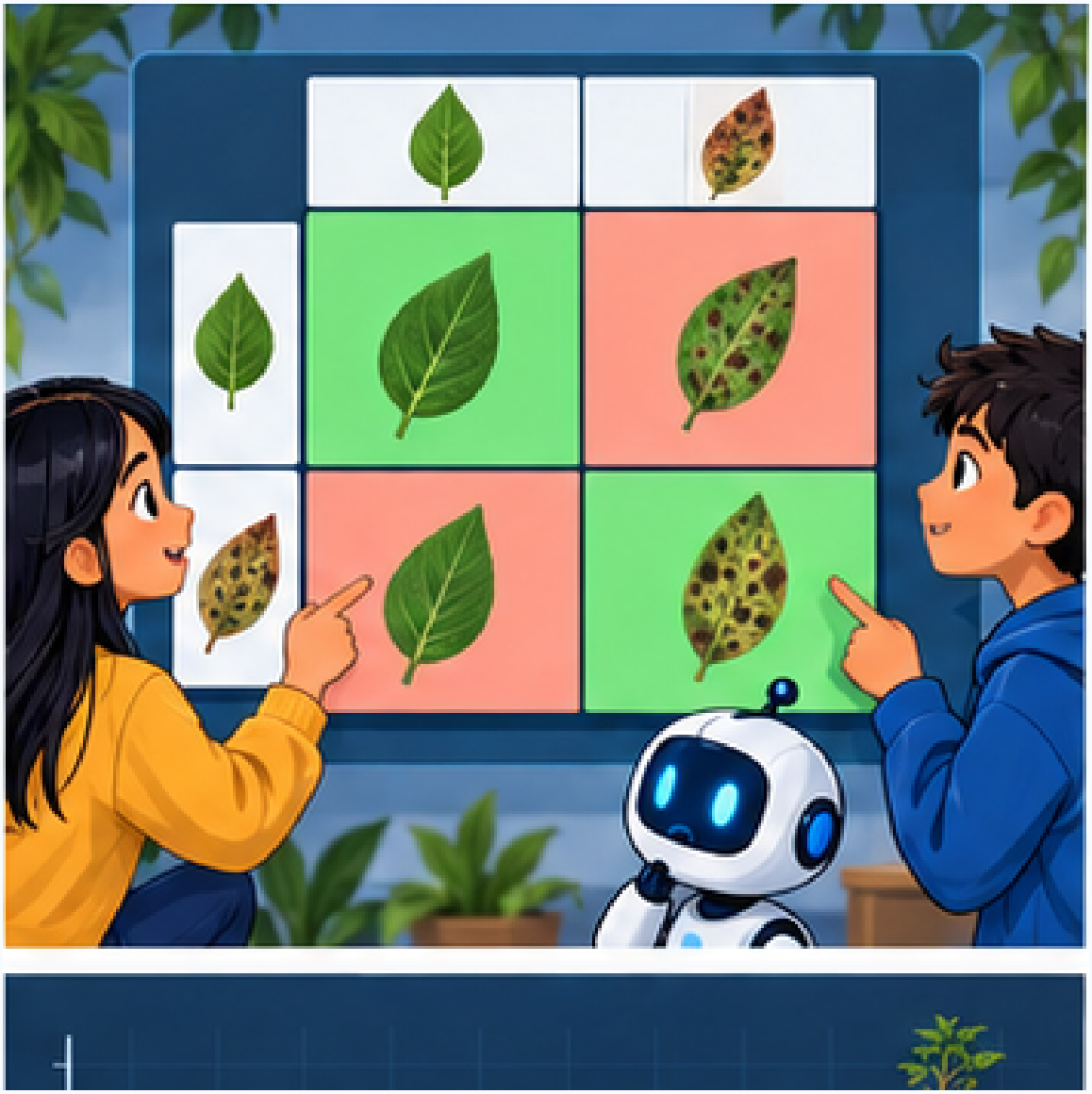
▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A **confusion matrix** separates **true positive**s, **false positive**s, **false negative**s, and **true negative**s so error types remain visible.



LEARNING TARGET Positive=spotted. TP=6,FN=2,FP=3,TN=9. Draw a matrix with actual labels in rows and **prediction**s in columns. Compute total and accuracy.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Declare the positive class and both axis orders

> Positive names the condition of interest, not a good outcome.

Matrix purpose and new fictional evaluation source

A **confusion matrix** shows which labels a classifier mixes up. It counts examples after **prediction**s are made; it does not fit or repair the classifier. Week 13 had reference/**prediction** pairs clear/clear, blocked/blocked, clear/blocked. A labelled table can preserve the type of that error as well as the total score.

This week uses eight new fictional evaluation records from a frozen classifier. They are not extra **prediction**s from the Week 13 code. Each ID appears once with a reviewed reference and one **prediction**. The grouped table below lists every ID. Keep reference and **prediction** paired by ID before counting.

Complete eight-record grouped table

Evaluation IDs	Reference	Prediction
V1 V2	clear	clear
V3	clear	blocked
V4 V5	blocked	clear
V6 V7 V8	blocked	blocked

Reference-row and prediction-column contract

Use reference labels for rows and predicted labels for columns. On both axes the order is clear, then blocked. Declare blocked as the **positive class** and clear as negative.

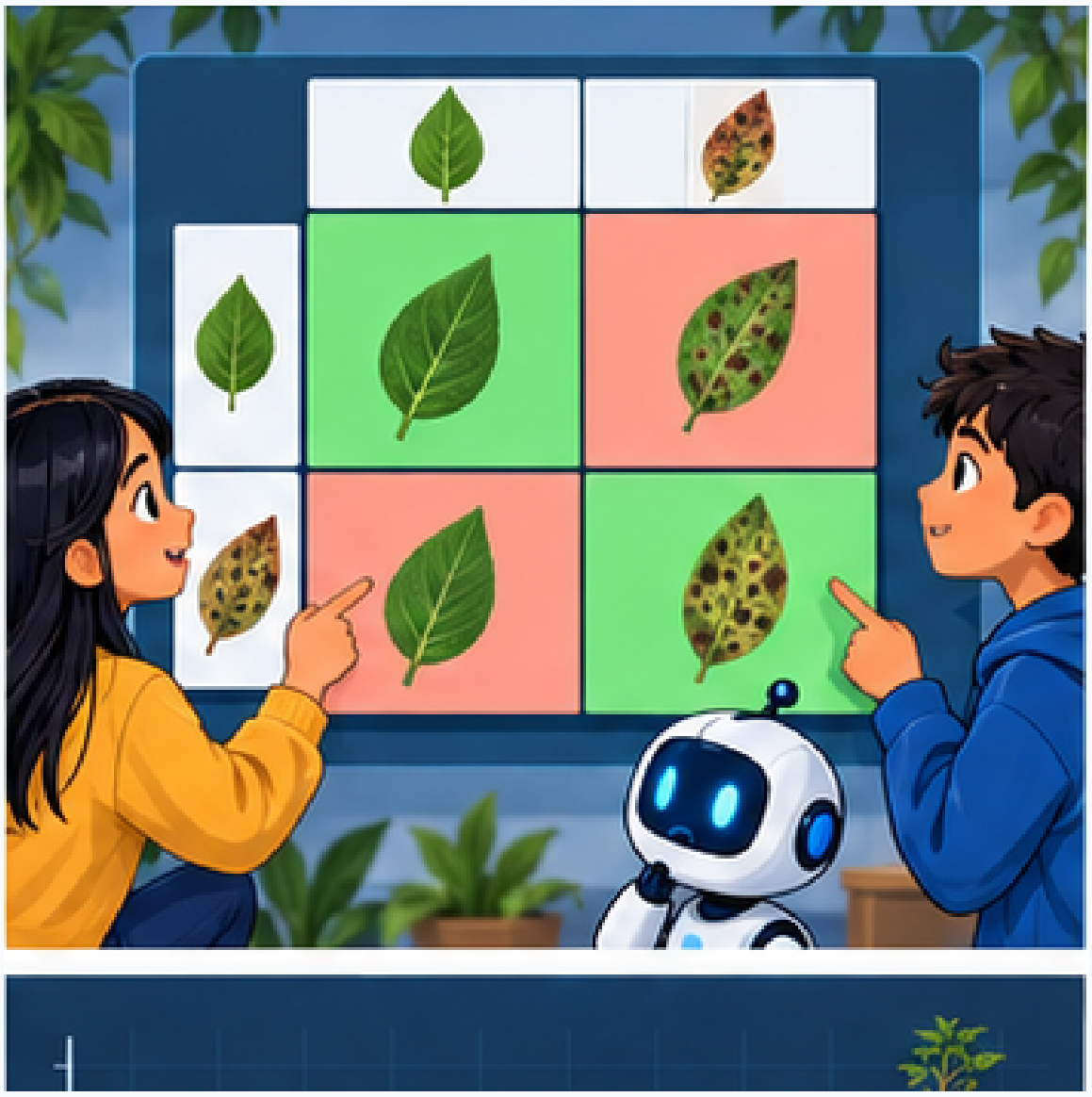
Positive means the condition of interest, not good or safe. A missed blocked case and a false alarm are different errors.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use **confusion matrix** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Put each reference/prediction pair in exactly one cell

> Axis labels make the numbers interpretable.

Complete eight-record counting program

```
pairs = [("V1", "clear", "clear"),
         ("V2", "clear", "clear"),
         ("V3", "clear", "blocked"),
         ("V4", "blocked", "clear"),
         ("V5", "blocked", "clear"),
         ("V6", "blocked", "blocked"),
         ("V7", "blocked", "blocked"),
         ("V8", "blocked", "blocked")]
labels = ["clear", "blocked"]
index = {label: i for i, label in enumerate(labels)}
counts = [[0, 0], [0, 0]]
for item_id, reference, prediction in pairs:
    counts[index[reference]][index[prediction]] += 1
print(counts)
correct = counts[0][0] + counts[1][1]
print(correct, len(pairs))
```

Trace IDs and assumptions

index maps clear to 0 and blocked to 1. Each paired record adds one to its reference row and **prediction** column. The script prints [[2, 1], [2, 3]] then 5 8. It assumes unique IDs and only these two labels.

Check IDs and missing or unresolved references before counting; the code does not validate all those conditions.

Read the labelled count matrix

Reference rows	Predicted clear	Predicted blocked
Reference clear	2	1
Reference blocked	2	3

Verify diagonal, errors and total

The diagonal cells count correct **prediction**s: 2+3=5. Off-diagonal cells count mistakes: 1+2=3. Every record must contribute exactly once. This is a count matrix, not percentages or probabilities.

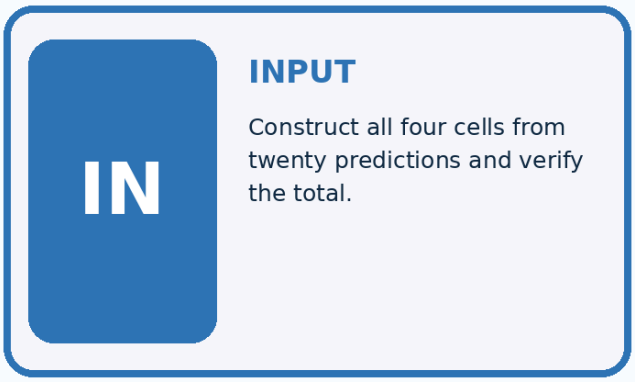
Keep the axis labels even when copying the numbers.

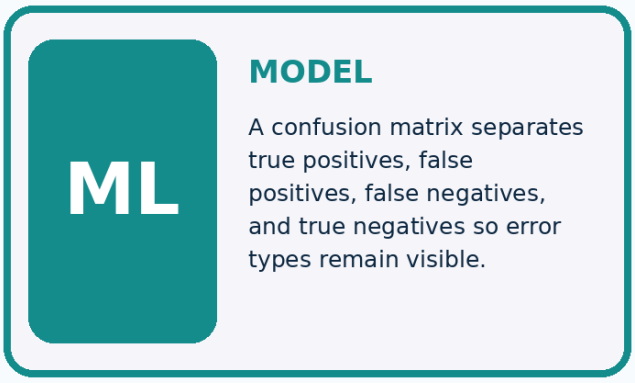
▼ Earlier wording, alternatives and picture transcripts

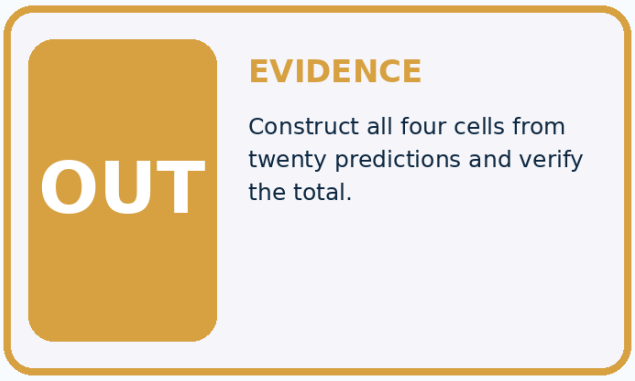
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Construct all four cells from twenty **prediction**s and verify the total.

MODEL IDEA A **confusion matrix** separates **true positive**s, **false positive**s, **false negative**s, and **true negative**s so error types remain visible.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD For spotted=positive: TP=6, FP=3, FN=2, TN=9. Check total: 6 + 3 + 2 + 9 = 20 cases.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

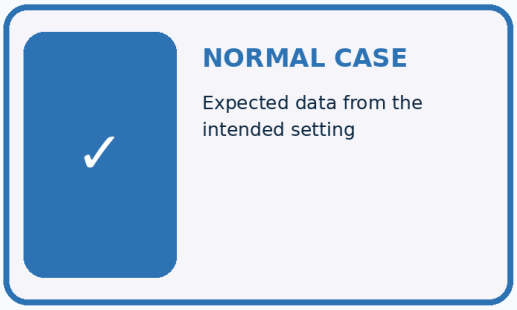
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

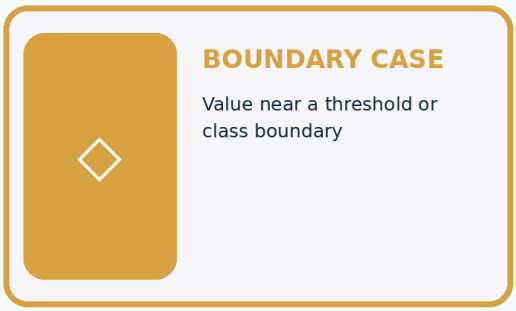
Compare two error types that one accuracy total hides

> A false alarm and a missed positive tell different stories.

Four evaluation conditions



NORMAL Expected data and expected behavior.

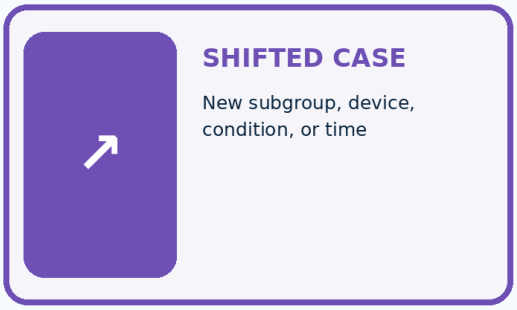


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: A review corrects one missed spotted leaf, moving it from FN to TP.

▼ **Check after predicting**

Prediction: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.

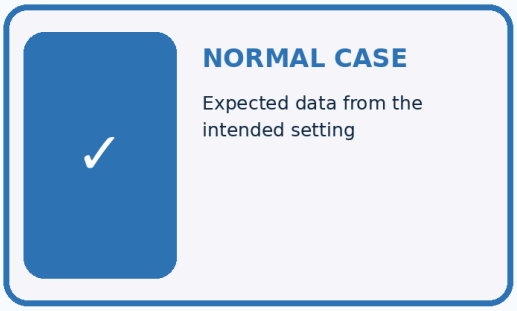
Reason: One decision becomes correct while the total number of evaluated leaves remains20.

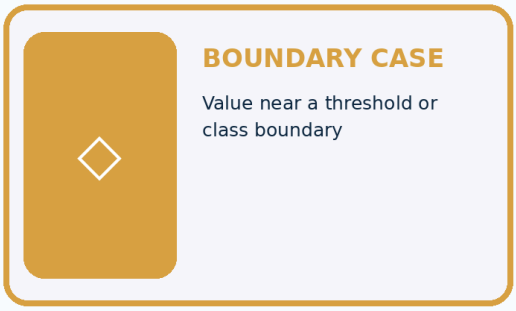
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

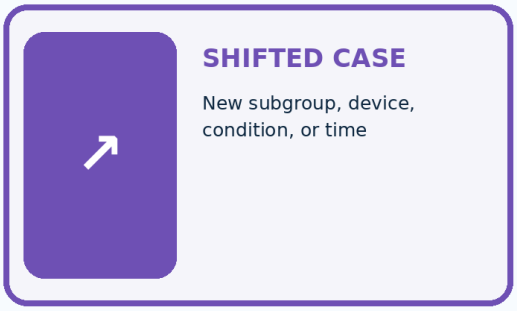
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read the four stories through their labelled cells

> Changing one decision moves one record; it does not add a new record.

Complete pictured facts and axis-order scope

Read the **confusion matrix** as four distinct stories about a plant decision.

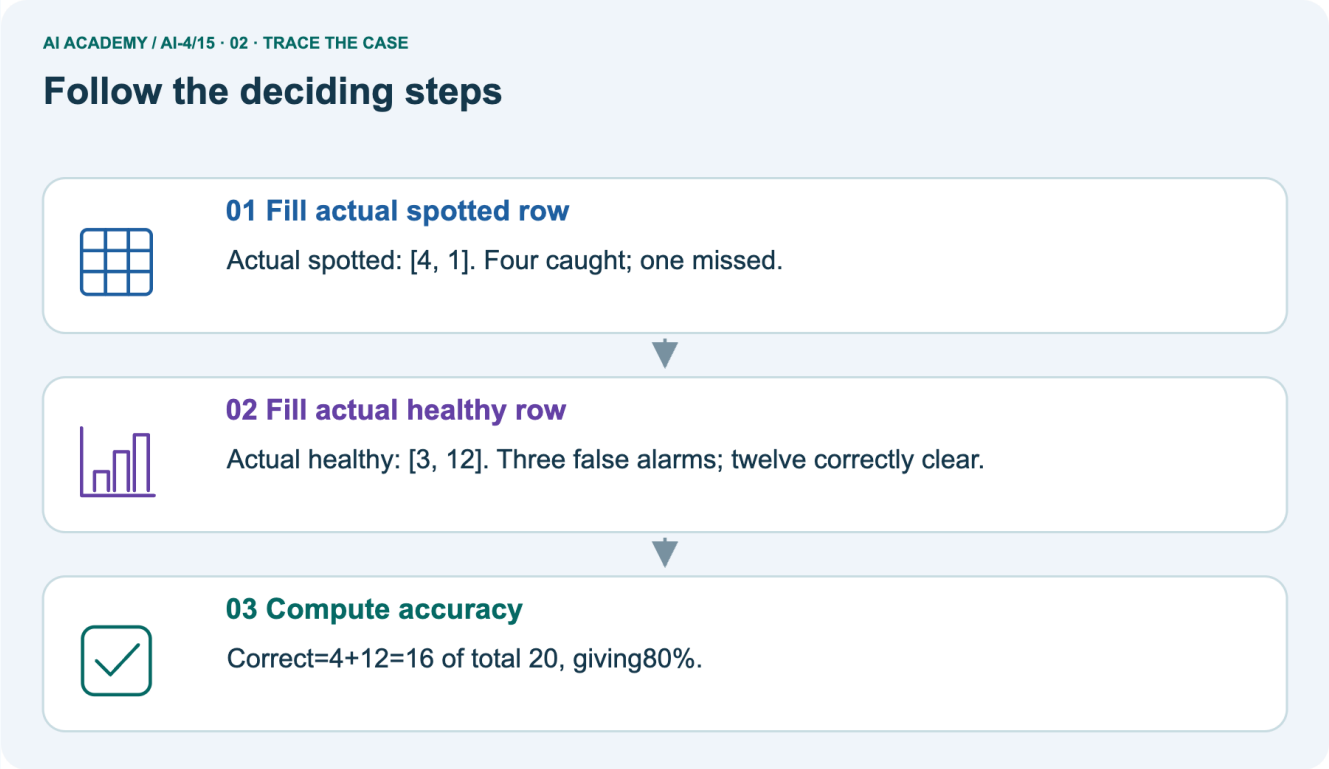
Positive means spotted. Toy results: TP=4, FN=1, FP=3, TN=12. Rows are actual spotted/healthy; columns predicted spotted/healthy.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main corridor matrix uses clear/blocked on both axes: [[2,1],[2,3]], positive blocked, total8,correct5. The pictured leaf matrix uses spotted/healthy on both axes: [[4,1],[3,12]], positive spotted,total20,correct16. Both use reference rows/prediction columns; the category order differs, so do not copy cell positions without labels.

In the pictured change, one output is corrected by review while its reference stays spotted: FN decreases1and TP increases1. Report this as a reviewed outcome, not improved untouched model accuracy; preserve the frozen model's original prediction and score.

Trace actual spotted and healthy rows



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

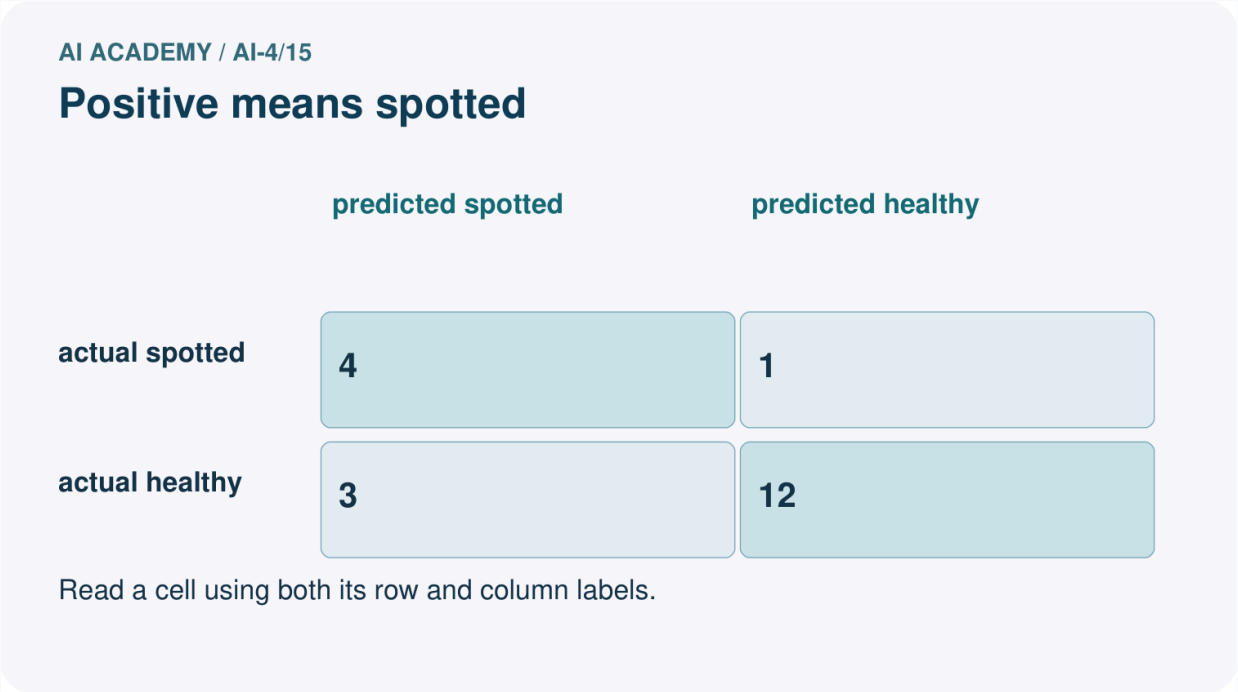
STEP	WHAT HAPPENS AND WHY
1. Fill actual spotted row	Actual spotted: [4, 1]. Four caught; one missed.
2. Fill actual healthy row	Actual healthy: [3, 12]. Three false alarms; twelve correctly clear.
3. Compute accuracy	Correct=4+12=16 of total 20, giving80%.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Correct=4+12=16 of total 20, giving**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Inspect the original labelled matrix

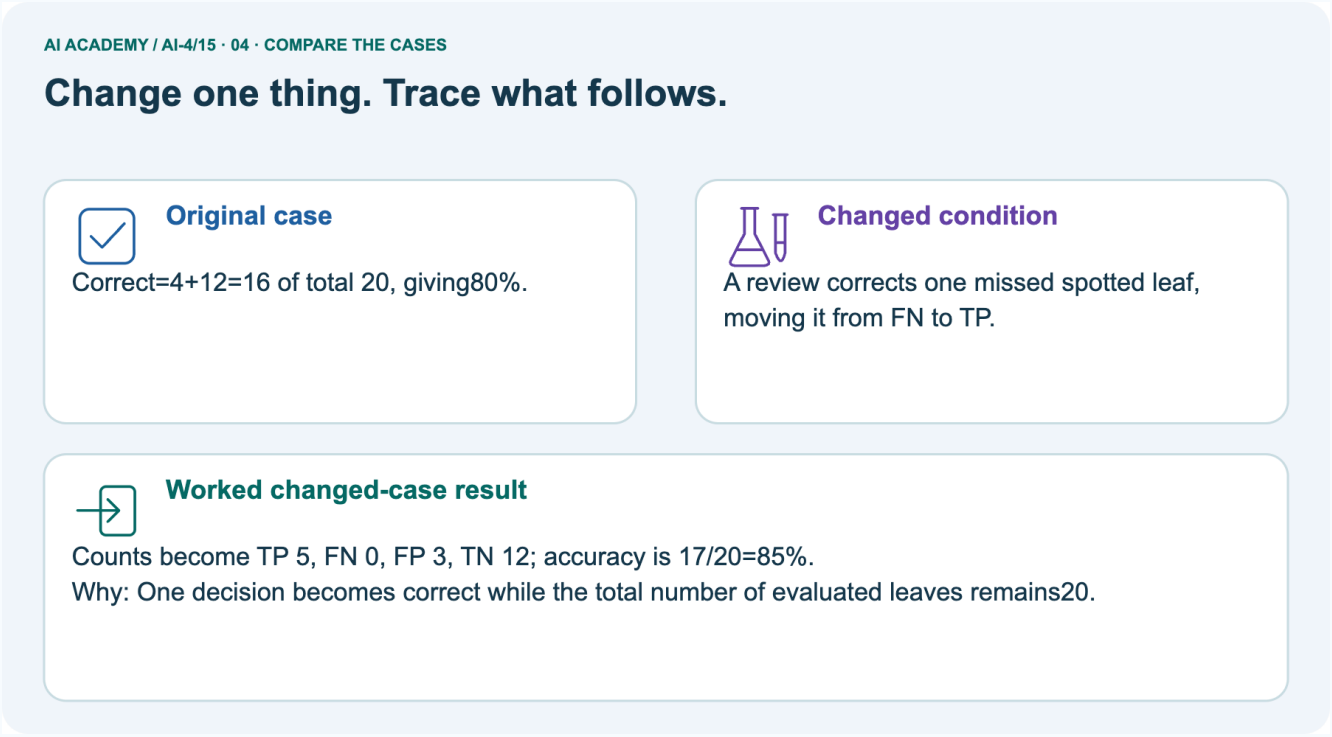


Follow the labelled picture with the matching explanation. Open the vector diagram

Predict a reviewed FN-to-TP decision change

- Change one thing: A review corrects one missed spotted leaf, moving it from FN to TP. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain limits and total preservation

- The matrix depends on positive-class definition and reference labels. Accuracy alone hides the difference between misses and false alarms.

- Read the labelled picture: Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to corridor V3/V4 error identities

Blocked is the positive class

REFERENCE AND PREDICTION	CELL NAME	IDS AND COUNT
clear then clear	True negative TN	V1 V2 count 2
clear then blocked	False positive FP	V3 count 1
blocked then clear	False negative FN	V4 V5 count 2
blocked then blocked	True positive TP	V6 V7 V8 count 3

Read reference first and prediction second. Here a false positive is a false blocked alarm; a false negative is a missed blocked case.

Place V3 and V4 in the matrix. Explain which is a false positive and which is a false negative when blocked is positive.

True means **prediction** agrees with the reference. It does not prove that the reference annotation is infallible. Resolve disputed labels with the documented review policy rather than silently changing them to match **prediction**s.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 15 • WORKED EVIDENCE

1. For spotted=positive: TP=6, FP=3, FN=2, TN=9.
2. Check total: 6 + 3 + 2 + 9 = 20 cases.
3. Correct predictions = TP + TN = 15; accuracy = 15 ÷ 20 = 75%.
4. Inspect false alarms and misses separately before choosing an improvement.

STEP 1 For spotted=positive: TP=6, FP=3, FN=2, TN=9.

STEP 2 Check total: 6 + 3 + 2 + 9 = 20 cases.

STEP 3 Correct **prediction**s = TP + TN = 15; accuracy = 15 ÷ 20 = 75%.

STEP 4 Inspect false alarms and misses separately before choosing an improvement.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

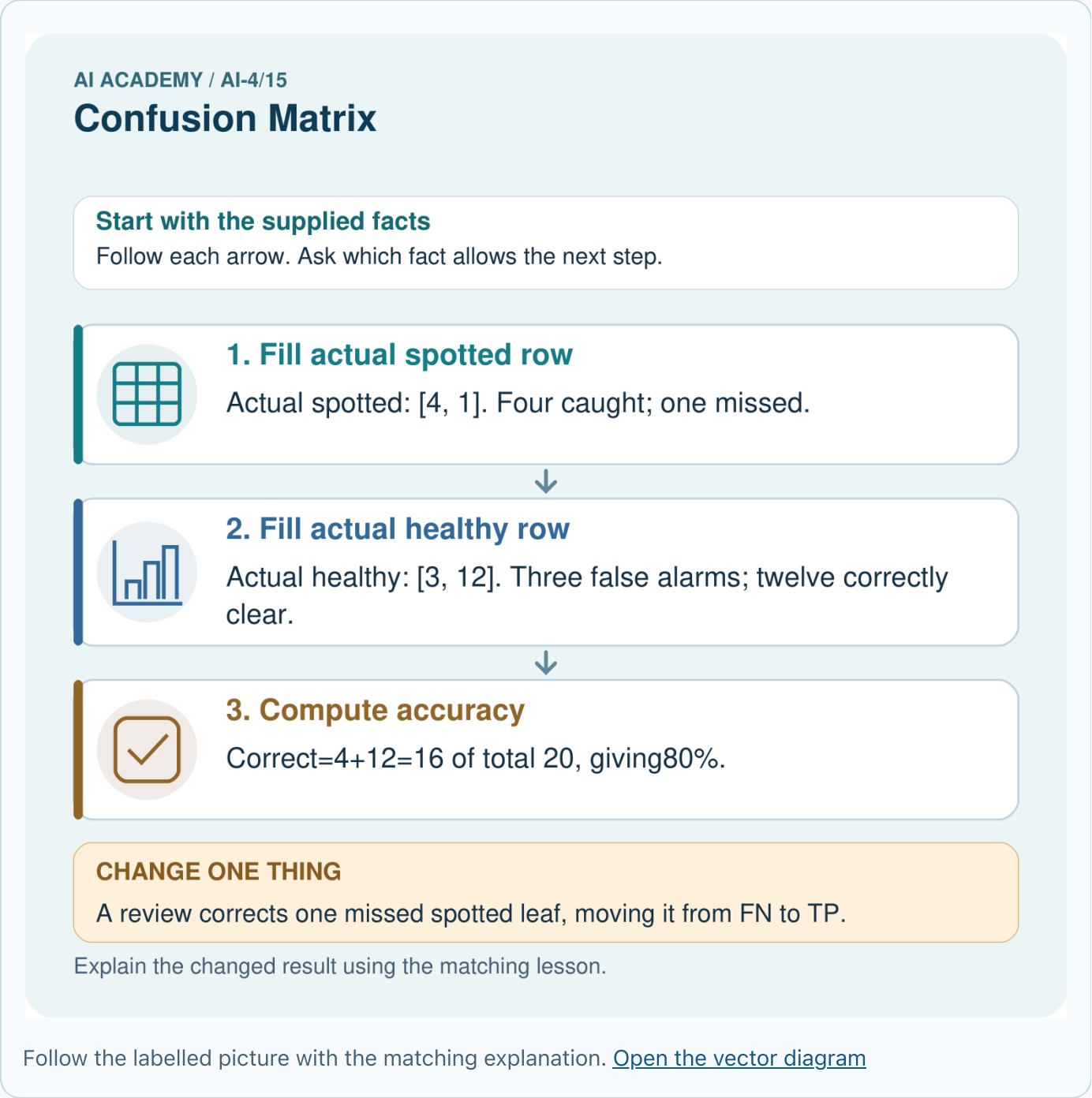
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Confusion Matrix

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a cell named without its class or axes

> Keep reference quality and excluded records visible.

Diagnose the main claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE The diagonal alone tells everything important about a classifier.

Check interpretation and evidence limits

- Always report the evaluated split, number of records, reference and **prediction** axes, label order and **positive class**. Reversing the axis order changes positions, not the underlying outcomes.

  Transposing rows and columns swaps the two kinds of error unless the labels are corrected. Read the headers before interpreting a diagram from another source.

- A zero cell means no such event in this evaluated set, not that it is impossible. A **confusion matrix** does not show the cause of errors, label quality or performance on future data. Keep the ID-level records for investigation.

  Next week examines accuracy when class counts are unequal.

> **Two classifiers have accuracy 5/8. One misses two blocked records and the other misses three. Explain how both statements can be true and what must accompany their matrices for a fair comparison.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: The diagonal alone tells everything important about a classifier. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

Practise error types and verify every marginal total

> Equal accuracy can conceal a different mix of mistakes.

Try W2 with the eight-record source

Use workbook W2 for the connected example “Confusion matrices”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Read row/column totals and alternative error pattern

Row totals are 3 reference-clear and 5 reference-blocked. Column totals are 4 predicted-clear and 4 predicted-blocked. All four cells sum to 8. Accuracy is the diagonal total divided by the total: 5/8, or 62.5 percent. The Week 13 three-record matrix was [[1,1],[0,1]] in this same axis order, giving 2/3.

Another classifier on these same eight reference-labelled records could have matrix [[3,0], [3,2]]. It also gets 5/8 correct, but has zero **false positive**s and three **false negative**s.

Our worked classifier has one **false positive** and two **false negative**s. Equal accuracy does not settle which error pattern is preferable; consider the task and **evidence** before choosing.

Challenge positive-means-good reasoning

- Use the worked case for Confusion matrices. Proposed explanation: “Positive means good”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured one-record move

- Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve the independent new counts

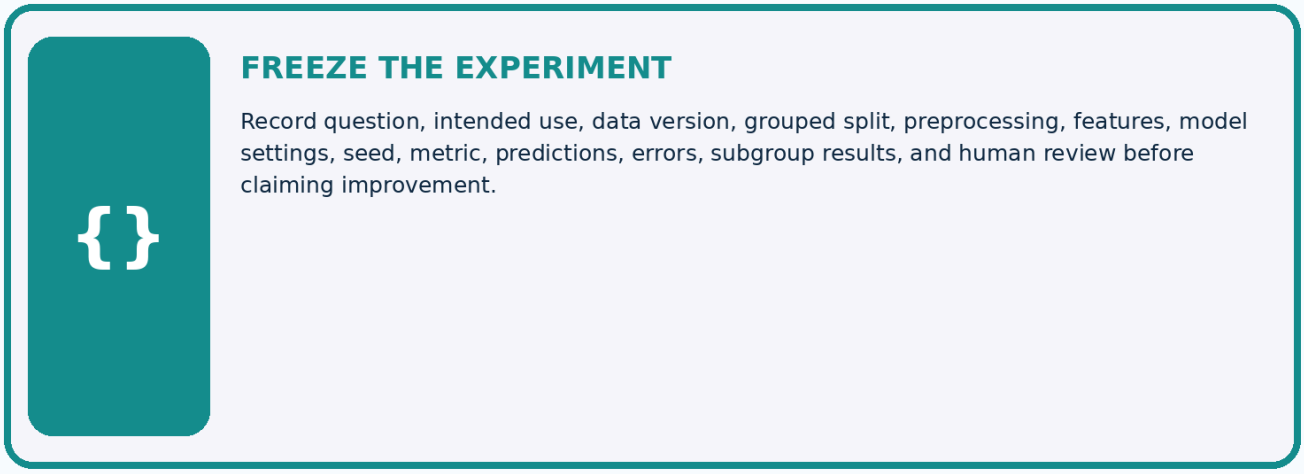
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + **positive class** + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Explain the two error types

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run Matrix Build with all twenty pairs supplied

> A count audit should account for every example exactly once.

Complete twenty fictional reference/prediction pairs

Case ID	Reference	Prediction
C01	spotted	spotted
C02	spotted	spotted
C03	spotted	spotted

Case ID	Reference	Prediction
C04	spotted	spotted
C05	spotted	healthy
C06	healthy	spotted
C07	healthy	spotted
C08	healthy	spotted
C09	healthy	healthy
C10	healthy	healthy
C11	healthy	healthy
C12	healthy	healthy
C13	healthy	healthy
C14	healthy	healthy
C15	healthy	healthy
C16	healthy	healthy
C17	healthy	healthy
C18	healthy	healthy
C19	healthy	healthy
C20	healthy	healthy

Declare axes and trace all totals

Matrix Build uses the twenty C01–C20 records, matching the original pictured counts. Declare spotted positive and use spotted/healthy on both reference-row and prediction-column axes. Each ID contributes once: C01–C04 TP, C05 FN, C06–C08 FP, C09–C20 TN.

Verify row totals5and15, column totals7and13, total20,correct16,accuracy80%.

This lab is separate from the eight-record corridor example and the independent new count set.

Normal, axis-order, missing-reference and positive-definition probes

Complete matrix probe	Expected bounded check
Normal C05 spotted/healthy	False negative, first reference row/second prediction column.
Boundary reverse both label orders to healthy/spotted	Same records become [[12,3],[1,4]], still16/20correct; labels must travel with cells.
Missing reference on an extra proposed C21	Cannot place an unknown reference as healthy; resolve or document exclusion separately, with its denominator.
Changed positive definition healthy, unchanged C01–C20 records	With original spotted/healthy axes unchanged: TP12,TN4,FP1,FN3; same numeric matrix and accuracy.

Count, verify and explain

1. Count each supplied C01–C20 card into the labelled matrix, then verify all totals independently before checking the optional complete program. Explain C05andC06in ordinary words rather than only abbreviations. Compare the axis-order and positive-definition probes separately: changing display order and changing the condition of interest have different effects.

   Retain original frozen predictions when recording reviewed decisions. No model is fitted or repaired by calculating the table. Keep the independent new TP/FN/FP/TN counts for an unaided first attempt.

Optional complete checked counting program

```
def build_counts(pairs, labels):
    if len(labels) != 2 or len(set(labels)) != 2:
        raise ValueError('Declare exactly two distinct labels in axis order')
```

```
    ids = [item_id for item_id, actual, predicted in pairs]
    if len(ids) != len(set(ids)):
        raise ValueError('Repeated evaluation ID: review the source')
    index = {label: i for i, label in enumerate(labels)}
    counts = [[0, 0], [0, 0]]
    for item_id, actual, predicted in pairs:
        if actual not in index or predicted not in index:
            raise ValueError('Missing, unresolved or unsupported label')
        counts[index[actual]][index[predicted]] += 1
    total = sum(sum(row) for row in counts)
    correct = counts[0][0] + counts[1][1]
    return counts, correct, total


# The complete printed C01–C20 fixture, with reference first in every pair.
pairs = [(f'C{i:02}', 'spotted', 'spotted') for i in range(1, 5)]
pairs += [('C05', 'spotted', 'healthy')]
pairs += [(f'C{i:02}', 'healthy', 'spotted') for i in range(6, 9)]
pairs += [(f'C{i:02}', 'healthy', 'healthy') for i in range(9, 21)]
labels = ['spotted', 'healthy']
counts, correct, total = build_counts(pairs, labels)
print(labels, counts)
print(correct, total, correct / total if total else 'unavailable')
# Empty evaluation has zero counts but no observed accuracy, never an invented 0%.
```

Retain evidence and actual execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

<details>
<summary>▼ Earlier wording, alternatives and picture transcripts</summary>

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.
</details>

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Matrix Build: convert twenty actual/predicted pairs into four cells, recompute totals, and explain each error in plain language.

1. Write the hypothesis and expected **evidence** before running.

2. Matrix Build: convert twenty actual/predicted pairs into four cells, recompute totals, and explain each error in plain language.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: build a matrix from fresh error counts

> State the class convention, verify the total and show the accuracy denominator.

Complete selected counts and scope

On20 cases: **true positive**s5, **false negative**s4, **false positive**s2, **true negative**s9. Build the **confusion matrix** with actual class as rows and predicted class as columns. Compute accuracy.

Declare actual positive/negative as row order and predicted positive/negative as column order before placing the supplied counts. If you illustrate positive as spotted, label that interpretation explicitly; the generic count task itself does not supply a diagnosis. Use the new20-case totals, not the pictured counts. No row-level IDs or subgroup evidence are provided, so do not invent them.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 15 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Treat fern as positive. Counts are TP=6, FP=2, FN=3 and TN=9. How many examples and how many correct **prediction**s are there?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Publish the Plant Classifier **confusion matrix** with positive-class definition, counts, denominator, and test-set version.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Positive=spotted. TP=6,FN=2,FP=3,TN=9. Draw a matrix with actual labels in rows and **prediction**s in columns. Compute total and accuracy.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's error stories beside its total score

> A labelled matrix can be checked against the underlying pair record.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Publish the Plant Classifier **confusion matrix** with positive-class definition, counts, denominator, and test-set version.

Save the model/data/label versions, frozen ID-linked predictions, reviewed references, positive definition, axis orders, matrix counts, totals and exclusions. If a human changes an output after review, keep that outcome separate from the original model-only score. A matrix reveals error types but does not verify reference truth, equal consequences or performance on new conditions.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

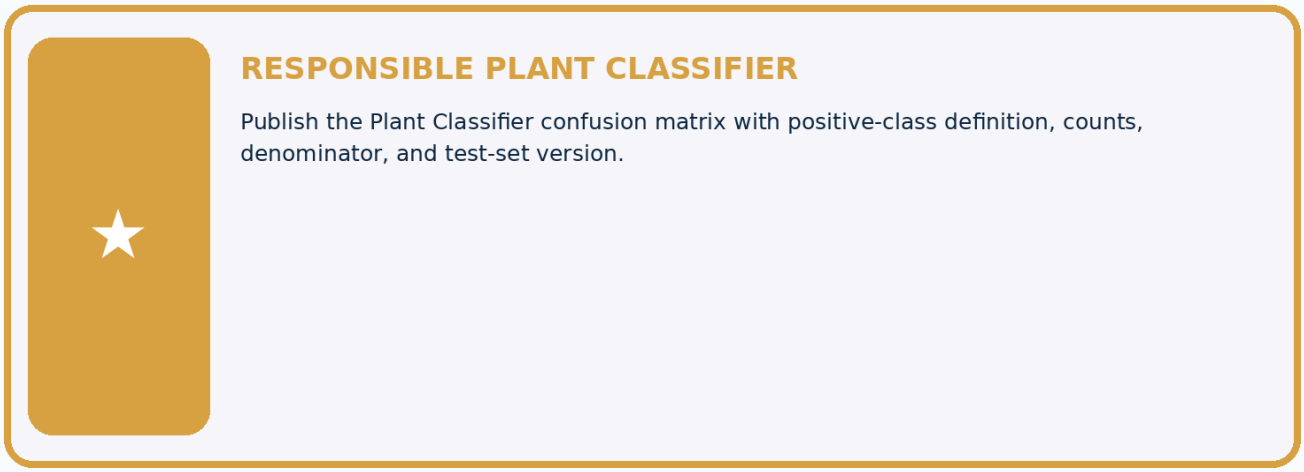
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with a labelled error and its denominator

> Explain why identical accuracies need not tell the same story.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Build a labelled **confusion matrix** from matched reference and **prediction** pairs

> **My deciding step, evidence and one remaining question**
>
> _____
>
> _____
>
> _____
>
> _____

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
true positive	positive case predicted positive	
false positive	negative case predicted positive	
false negative	positive case predicted negative	
true negative	negative case predicted negative	

Closer-explanation terms

Word	Meaning
Confusion matrix	A table counting reference and prediction combinations
Positive class	The class chosen as the condition of interest
False positive	A positive prediction when the reference is negative

Word	Meaning
False negative	A negative prediction when the reference is positive

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

VOCABULARY

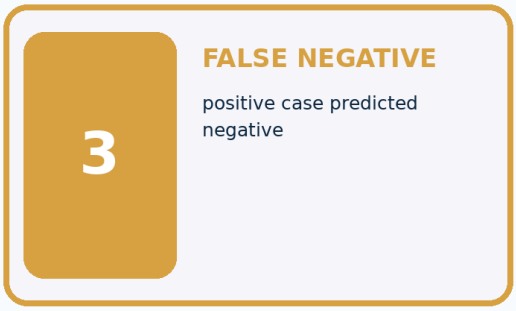
Words You Can Calculate and Defend

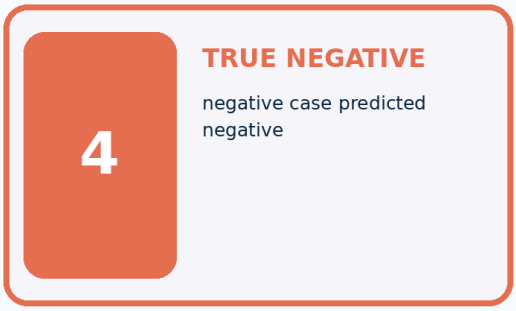
A definition matters when it lets you interpret a new dataset, model, or result.





</details>





MISCONCEPTION TO REPAIR The diagonal alone tells everything important about a classifier.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **true positive**, **false positive**, **false negative**, **true negative**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE The diagonal alone tells everything important about a classifier.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

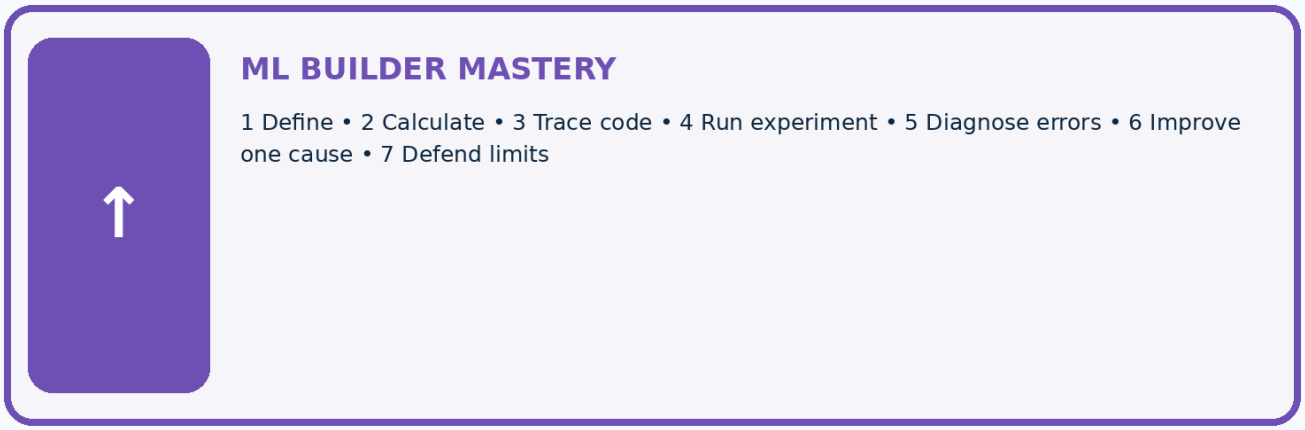
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh TP5/FN4/FP2/TN9 count task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

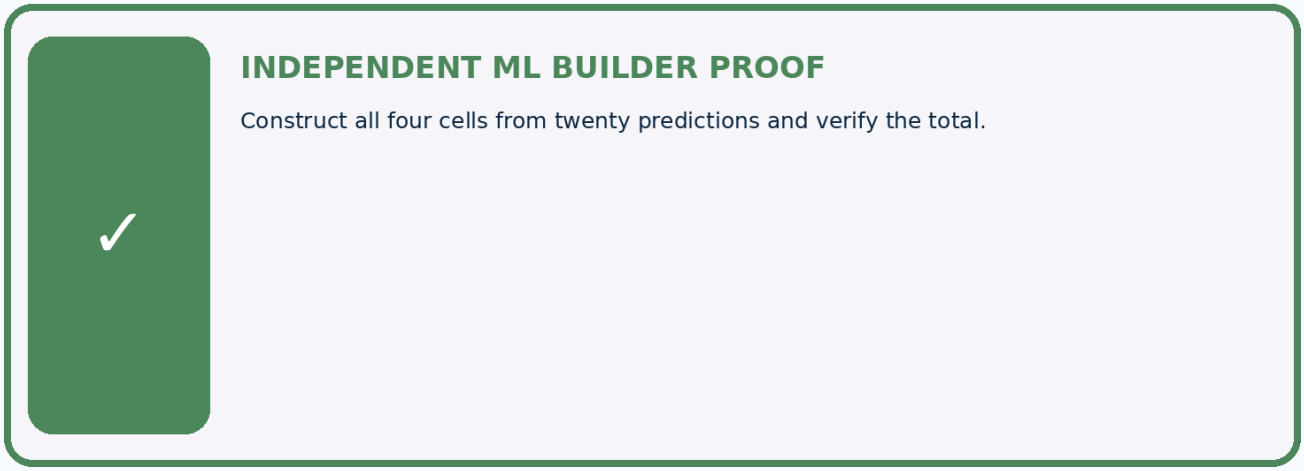
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Construct all four cells from twenty **prediction**s and verify the total.

NEXT BRIDGE Carry the experiment **evidence** into: Accuracy and Imbalance.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Positive is spotted. TP=8, FP=2, FN=4, TN=6.

Build actual-row/predicted-column matrix and total.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Actual spotted row is 8 spotted,4 healthy; actual healthy row is 2 spotted,6 healthy. Total=20; correct=14. State axes and **positive class**.

Evidence or stage	Value or result
True positive	8
False negative	4
False positive	2
True negative	6

▼ Earlier wording, alternatives and picture transcripts

Week 15 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Actual rows, predicted columns

Positive is spotted. TP=8, FP=2, FN=4, TN=6.

Build actual-row/predicted-column matrix and total.

Actual spotted row is 8 spotted,4 healthy; actual healthy row is 2 spotted,6 healthy. Total=20; correct=14. State axes and **positive class**.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Build a labelled confusion matrix from matched reference and prediction pairs	Explain the deciding step, show a trace or test, and state one limit.
Distinguish false positives from false negatives using a declared positive class	Explain the deciding step, show a trace or test, and state one limit.
Check matrix totals and explain why equal accuracy can hide different errors	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Read the **confusion matrix** as four distinct stories about a plant decision.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Read the **confusion matrix** as four distinct stories about a plant decision.

- Which labelled step matches this detail? Correct=4+12=16 of total 20, giving**80%**.
- 1. Compute accuracy
- 2. Fill actual spotted row
- 3. Fill actual healthy row
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compute accuracy Correct=4+12=16 of total 20, giving**80%**. Read the picture in order. Correct=4+12=16 of total 20, giving**80%**.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: A review corrects one missed spotted leaf, moving it from FN to TP. Which conclusion is supported by the supplied facts?
- 1. Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.

- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%. One decision becomes correct while the total number of evaluated leaves remains20.

Repair and keep the record

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score mean every such leaf is spotted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why is the k=1 outcome at length5 not uniquely determined without a tie rule?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

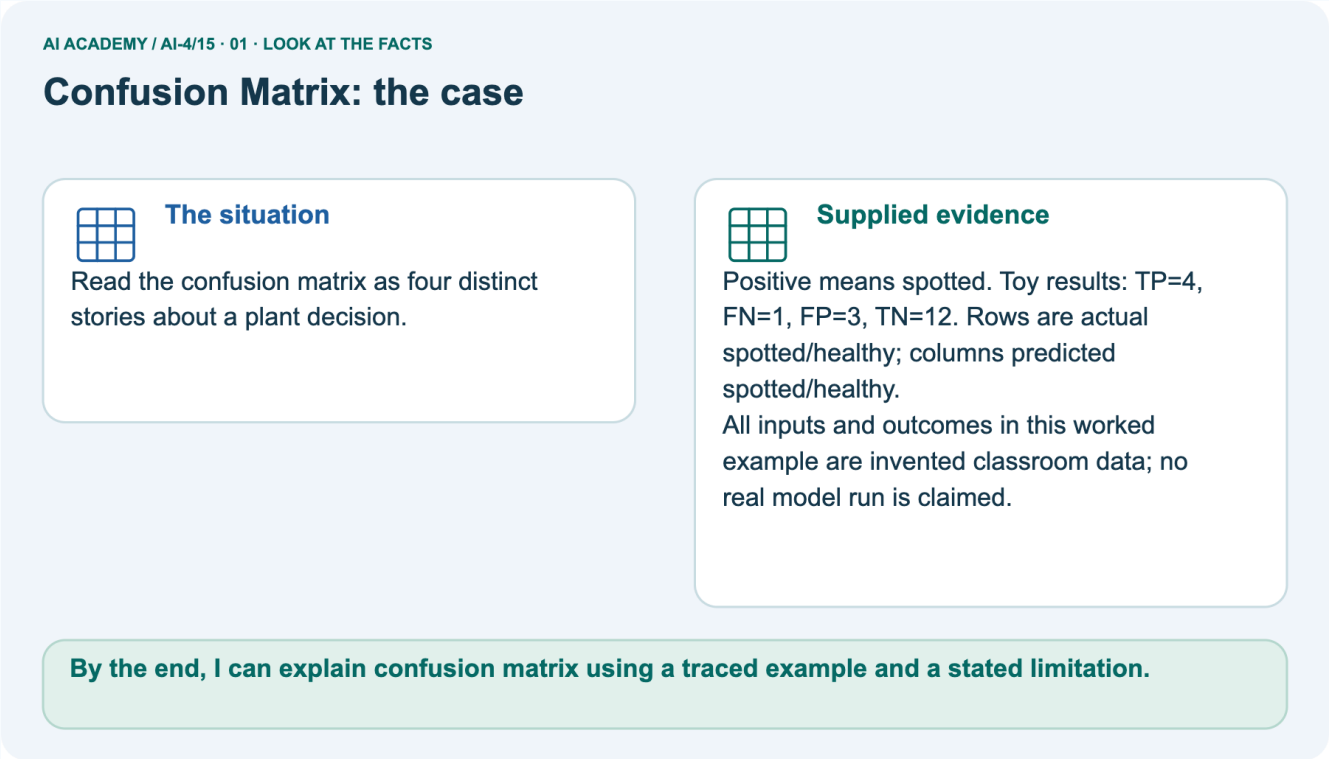
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Confusion Matrix: the case



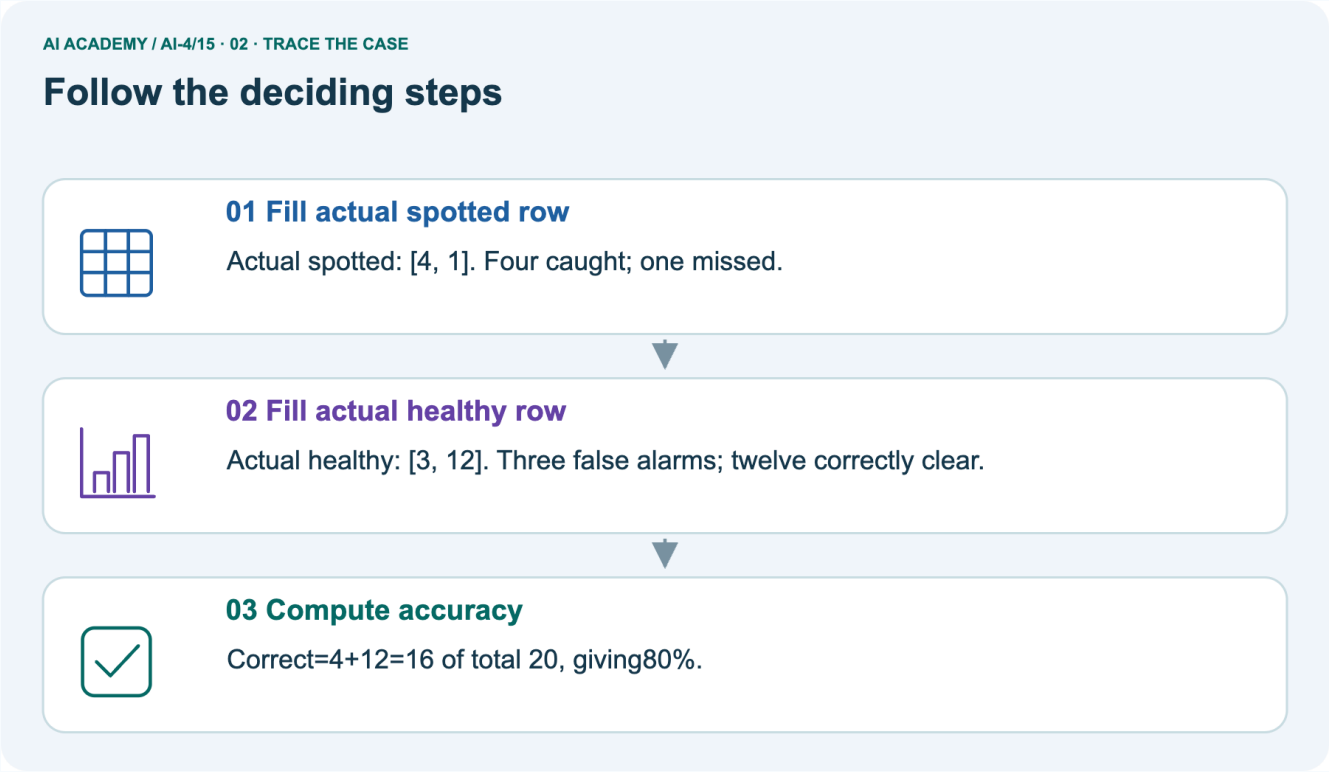
Follow the labelled picture with the matching explanation. Open the vector diagram

Read the **confusion matrix** as four distinct stories about a plant decision. Supplied facts: Positive means spotted. Toy results: TP=4, FN=1, FP=3, TN=12. Rows are actual spotted/healthy; columns predicted spotted/healthy.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



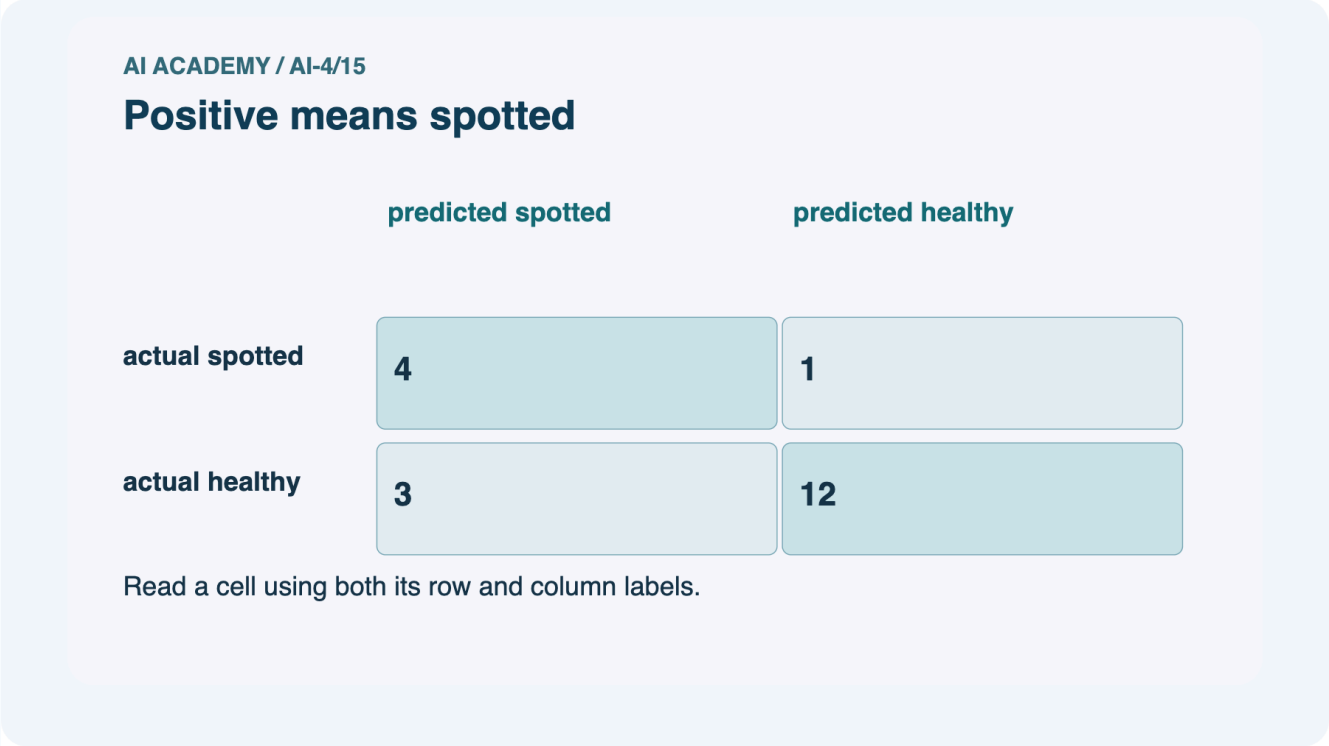
Follow the labelled picture with the matching explanation. Open the vector diagram

Fill actual spotted row: Actual spotted: [4, 1]. Four caught; one missed. → Fill actual healthy row: Actual healthy: [3, 12].

Three false alarms; twelve correctly clear. → Compute accuracy: Correct=4+12=16 of total 20, giving**80%**.

Point to each numbered stage and explain which recorded **input** it uses.

03. Positive means spotted

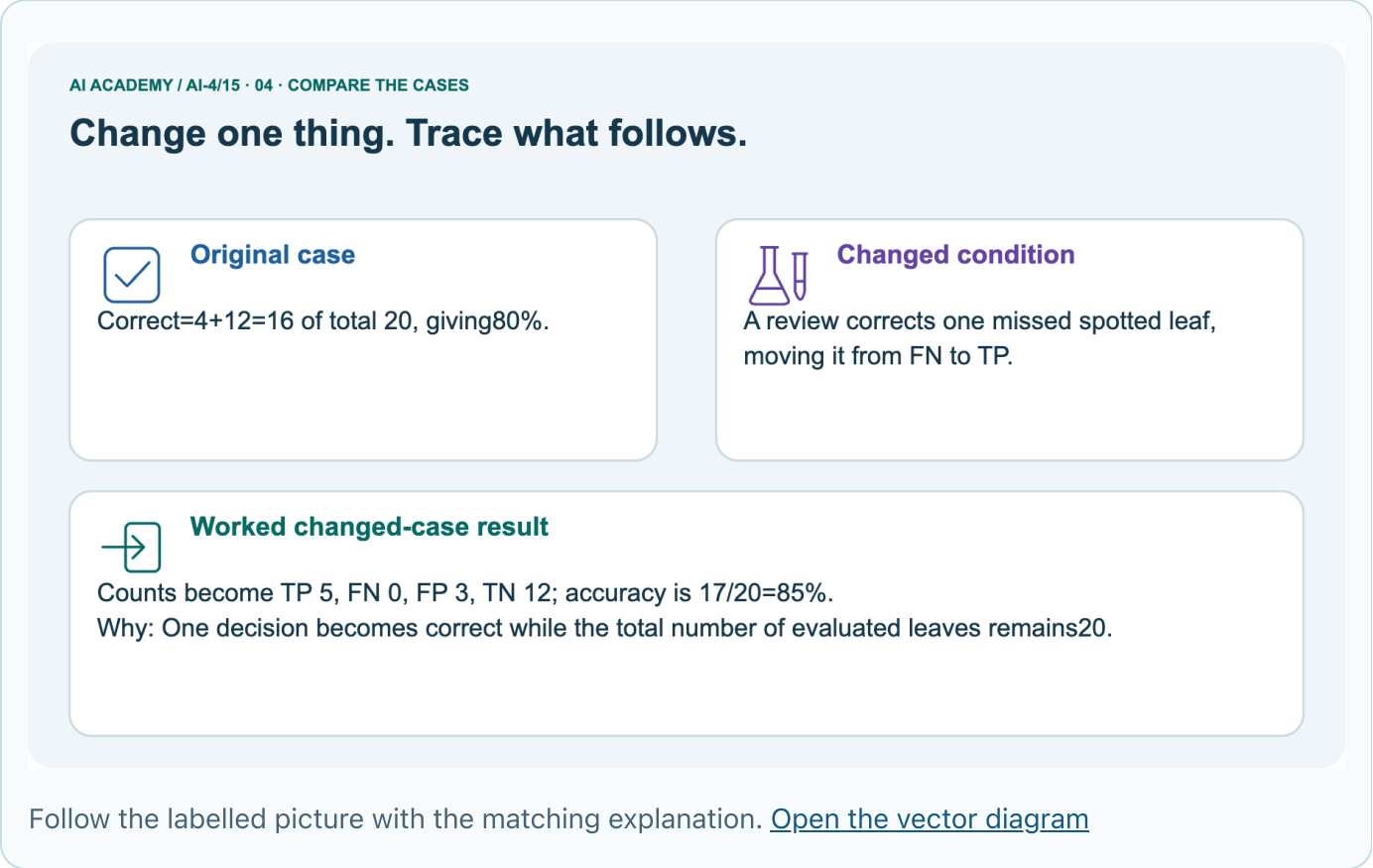


Follow the labelled picture with the matching explanation. Open the vector diagram

Positive means spotted. actual spotted: predicted spotted = 4, predicted healthy = 1. actual healthy: predicted spotted = 3, predicted healthy = 12.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



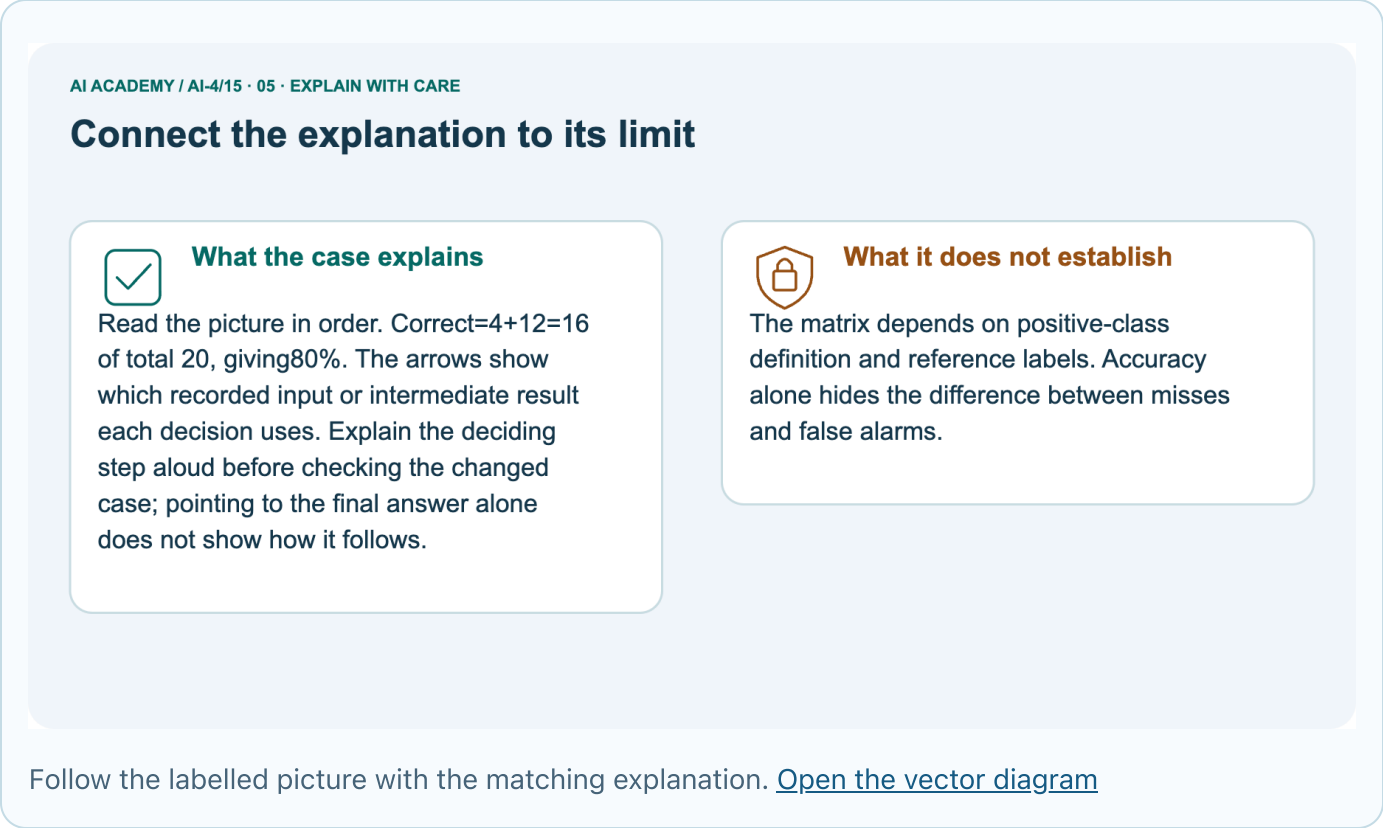
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Correct=4+12=16 of total 20, giving**80%**. Changed condition: A review corrects one missed spotted leaf, moving it from FN to TP. Expected result: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.

Reason: One decision becomes correct while the total number of evaluated leaves remains20.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

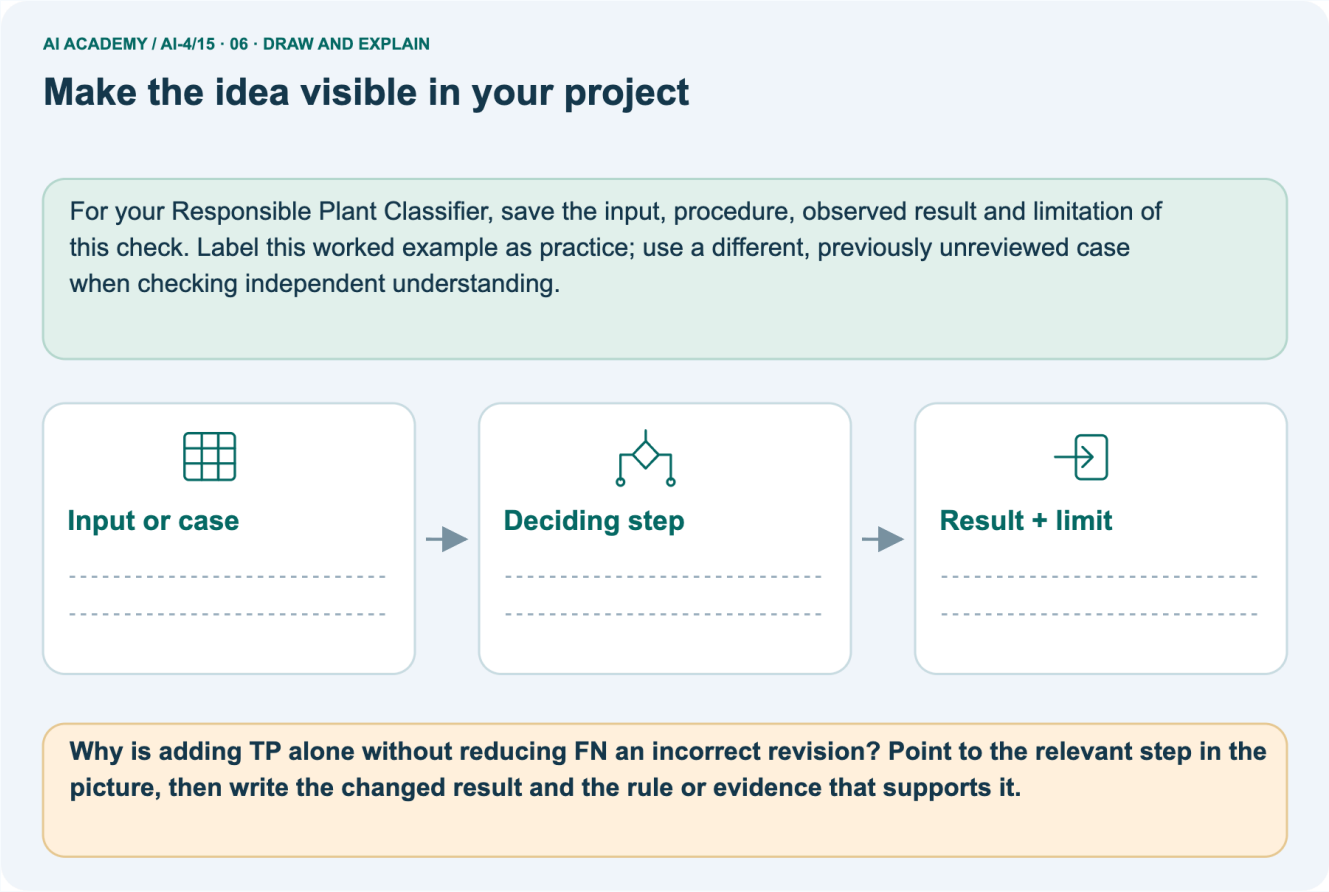
Read the picture in order. Correct=4+12=16 of total 20, giving**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: The matrix depends on positive-class definition and reference labels.

Accuracy alone hides the difference between misses and false alarms.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Confusion Matrix: the case

02. Follow the deciding steps

03. Positive means spotted

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.